

Understanding Data Quality Issues

Exploring the root causes of incomplete data and their impact on business decisions

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Why Data Goes Missing

Data incompleteness arises from multiple sources, each requiring different mitigation strategies. Understanding these patterns helps organizations build more resilient data systems.



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Common Causes of Incomplete Data



Unavailable Attributes

Key customer information may not exist or be accessible at transaction time, creating gaps in customer profiles.



Overlooked Importance

Data not deemed critical during initial system design may later prove essential for analytics.



Human Error

Misunderstandings during data entry or equipment malfunctions can prevent proper recording.



Consistency Checks

Inconsistent records may be deleted during validation, removing potentially valuable data.

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The Quality Paradox

Same Data, Different Views

Data quality is inherently subjective, varying by user needs and use cases.

Example Scenario

80% accurate addresses may satisfy marketing requirements but frustrate sales operations.



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Perspective Shapes Quality

Marketing Analyst

Views large customer database as high quality for target marketing. 80% address accuracy provides sufficient reach for campaigns.

Sales Manager

Considers same database inaccurate when trying to reach individual customers. Missing or outdated addresses directly impact revenue.

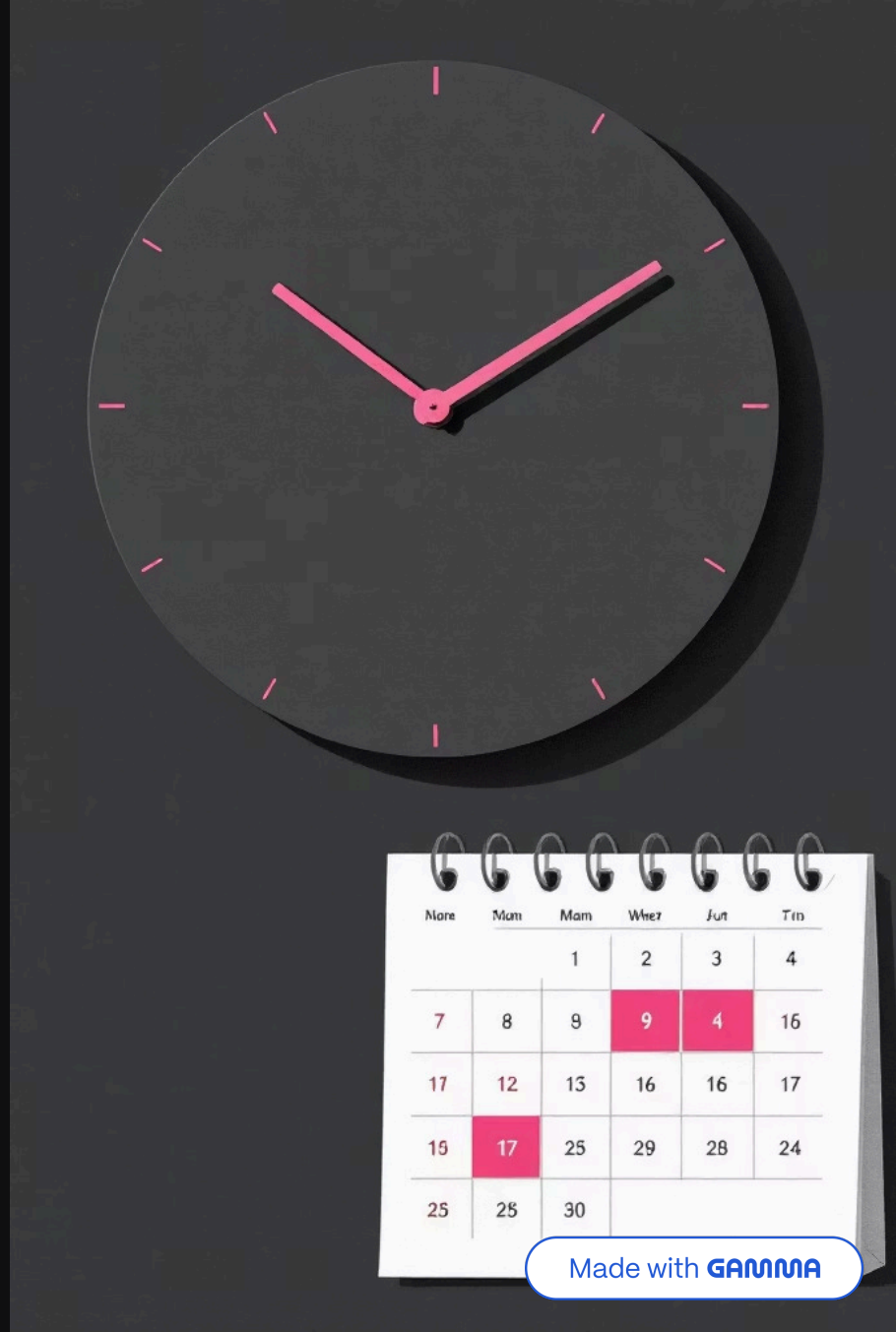
Data quality depends entirely on intended use and user expectations

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The Timeliness Factor

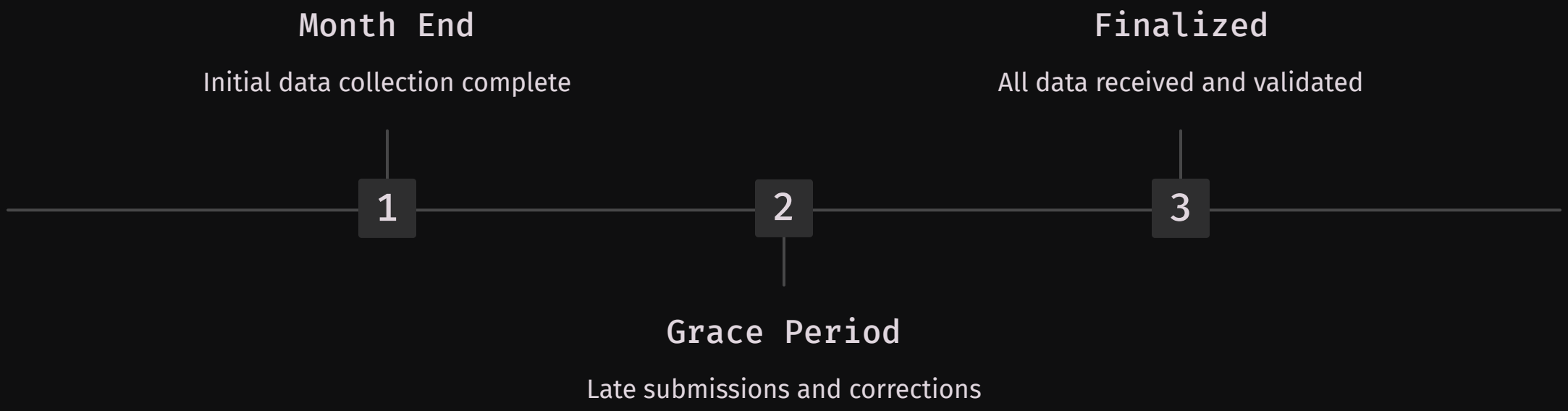
Monthly Sales Bonuses at AllElectronics

When sales representatives delay submissions or corrections arrive after month-end, database becomes temporarily incomplete. Quality degrades during transitional periods despite eventual accuracy.



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Timeliness Impact Timeline



During grace period, data quality suffers from incompleteness despite eventual correctness.

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Trust and Comprehension Matter



Believability

How much users trust the data based on past experiences with errors and corrections.



Interpretability

How easily users understand data structures, codes, and terminology used in the database.

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The Trust Problem

Historical Errors Create Lasting Damage

Even after all database errors are corrected, users remember past problems. Sales department continues distrusting data despite current accuracy, completeness, and consistency.

Accounting codes remain incomprehensible to sales teams, compounding the issue.

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Key Takeaways

1 Data quality is use-dependent

Different users assess the same database differently based on their specific needs.

2 Timeliness affects quality

Even correct data loses value when not available when needed.

3 Trust and clarity are essential

Accuracy alone doesn't guarantee quality—users must believe in and understand the data.

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